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Welcome to SecurCube

DIGITAL FORENSICS ANALYTICS

Technologies to simplify and manage the complexity of mobile networks for the better good against crime.

We know mobile communication: generate strong, validated and complete digital forensics investigations.

Complex mobile network data become a valuable and clear source of evidence.

Securcube Webinar Series

Our Company

SecurCube is a team of digital forensics experts and IT developers focused on justice.

Hands on experience and constant research:

highly specialized but easy to use technology for the successful investigation of **phone records** and the analysis of **cell networks**.

Correlate with other multiple sources to create a full circle data validation process.

We provide the method and the tools:
the best understanding of digital evidence and the clearest results for the courtroom.

Collect, organize, analyze, **validate**, and present your evidence with confidence.

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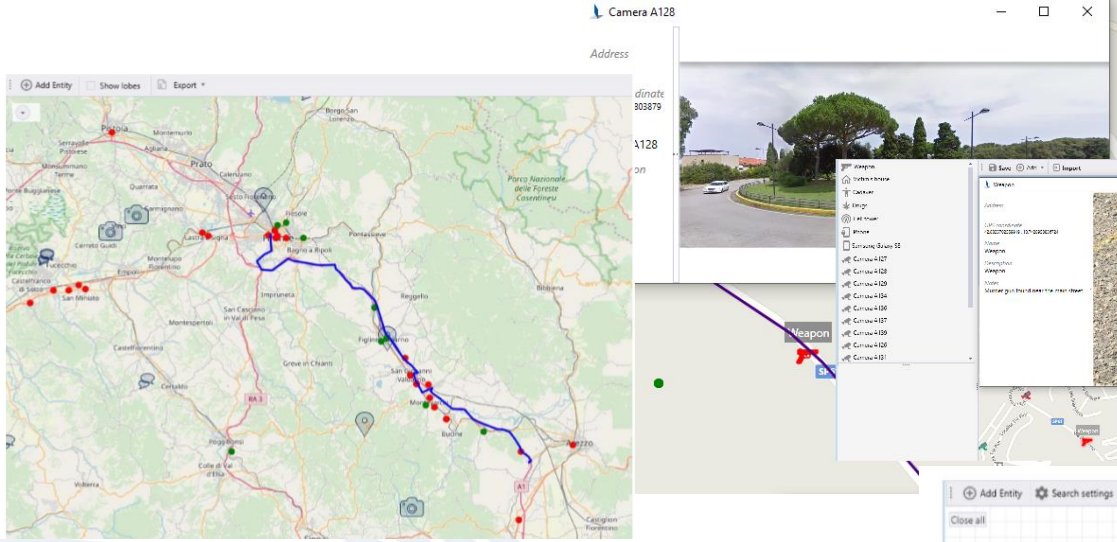
CDR investigation
Cell forensics

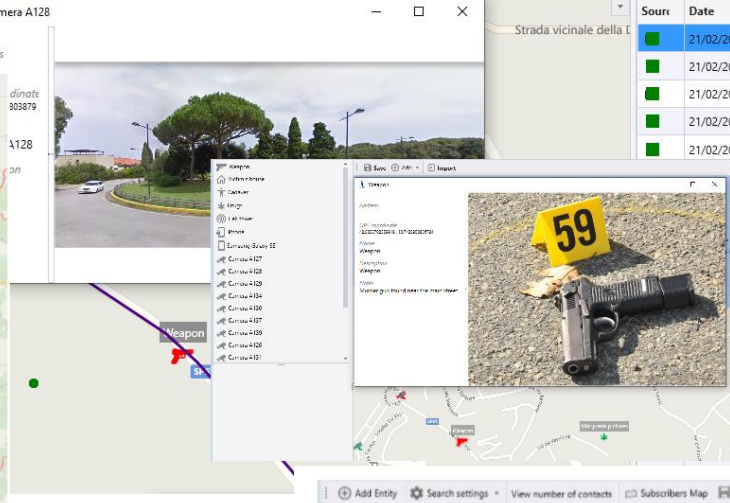
SecurCube Technology

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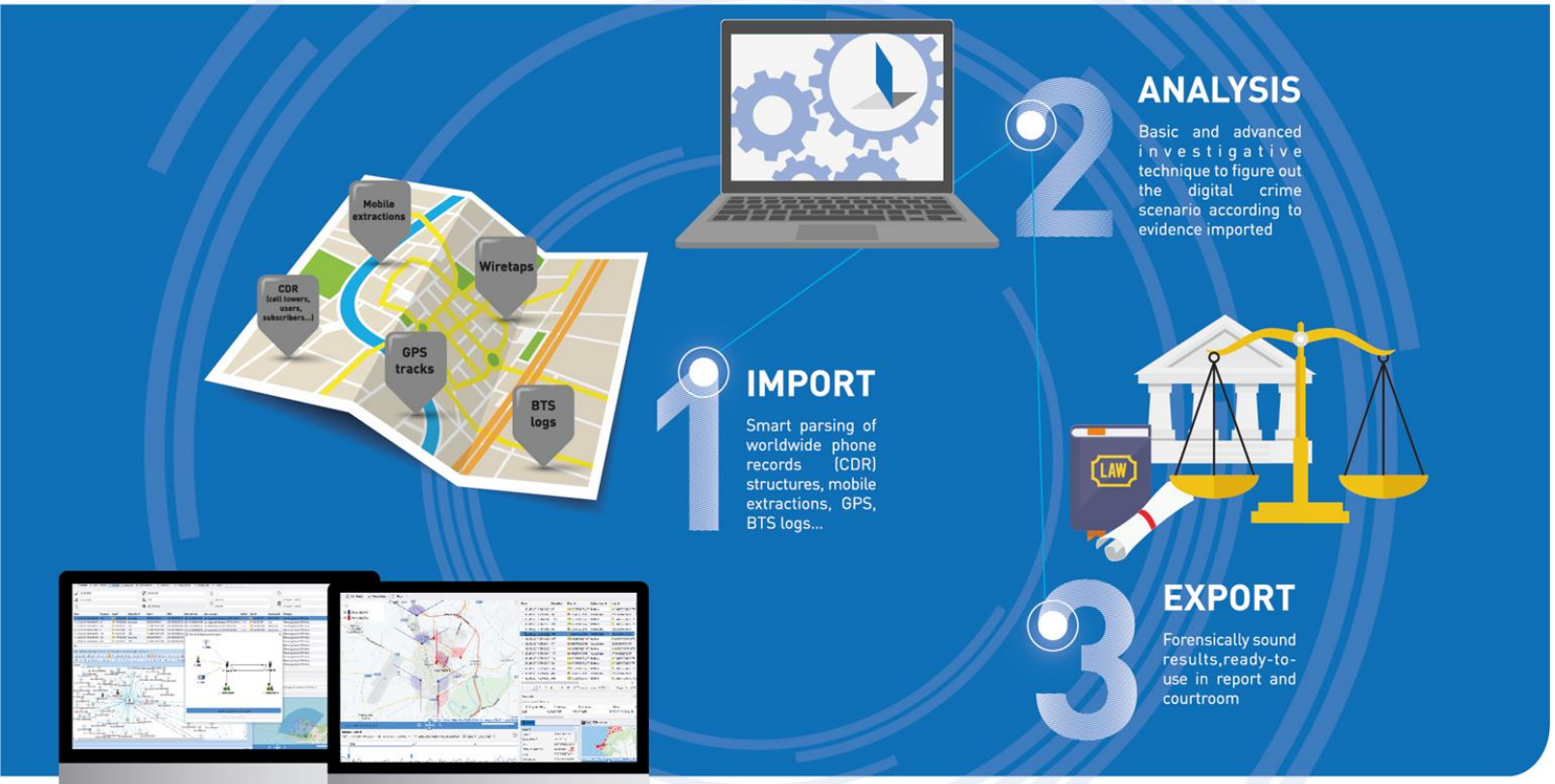
CDR investigation
SecurCube®PhoneLog

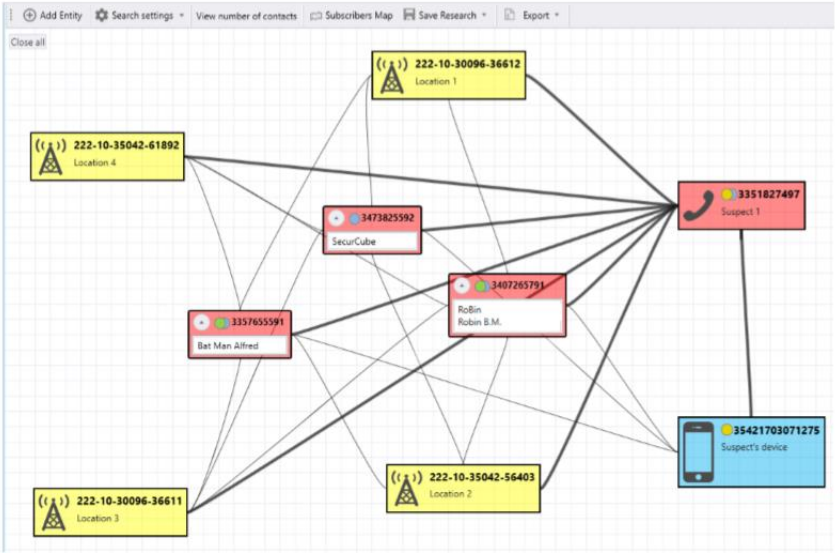
PhoneLog
YOU ARE REALLY ON TOP OF THINGS





Sourc	Date	Duration	User A	Subscriber A	IMEI A
	21/02/2015 08:23:30	596	3351827497	Bat Man | RoBin BTAMNA39E07Z404H	354217030712
	21/02/2015 08:34:04	186	3351827497	Bat Man | RoBin BTAMNA39E07Z404H	354217030712
	21/02/2015 08:38:09	34	3351827497	Bat Man | RoBin BTAMNA39E07Z404H	354217030712
	21/02/2015 13:17:13	65	3351827497	Bat Man | RoBin BTAMNA39E07Z404H	354217030712
	21/02/2015 13:20:49	61	3351827497	Bat Man | RoBin BTAMNA39E07Z404H	354217030712
	15 17:01:42	260	3351827497	Bat Man | RoBin BTAMNA39E07Z404H	354217030712
	15 17:06:27	7	3351827497	Bat Man | RoBin BTAMNA39E07Z404H	354217030712
	15 17:07:42	102	3351827497	Bat Man | RoBin BTAMNA39E07Z404H	354217030712
	15 07:51:12	55	3351827497	Bat Man | RoBin BTAMNA39E07Z404H	354217030712
	15 07:53:53	208	3351827497	Bat Man | RoBin BTAMNA39E07Z404H	354217030712
	15 07:59:44	231	3351827497	Bat Man | RoBin BTAMNA39E07Z404H	354217030712
	15 08:04:42	236	3351827497	Bat Man | RoBin BTAMNA39E07Z404H	354217030712
	15 08:08:57	197	3351827497	Bat Man | RoBin BTAMNA39E07Z404H	354217030712
	15 08:38:19	256	3351827497	Bat Man | RoBin BTAMNA39E07Z404H	354217030712
	15 08:44:40	229	3351827497	Bat Man | RoBin BTAMNA39E07Z404H	354217030712
	15 08:51:51	144	3351827497	Bat Man | RoBin BTAMNA39E07Z404H	354217030712





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Electronic devices & investigation

Today everyone has at least one device. It means a daily increasing amount of telecommunication exchanges. A valuable source of analysis for digital experts.

In particular:

- ✔ The device and the content
  (IMEI, IMSI, messages, photos, positions etc.)
- ✔ Cell Site Analysis – CSA
  (Call Detail Record, Historical Cell Site Location Information)



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The Basics:

Call Detail Record Analysis

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General definition – what is a CDR?

CDR (Call Detail Record) is a file, produced by the cellular providers (the carriers), containing information about telephone exchanges (e.g. calls). The record contains various attributes such as the date, time, duration, completion status, source number and destination number, amount of data transmitted or received, accordingly to the requested period.

- ✔ For billing reasons, carriers keep users' phone activities
- ✔ Digital experts can take advantage using the stored information for investigation purposes, reconstructing the suspect's movements and habits, junctions with other potential involved.



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The importance of CDR in digital forensics

A device content can tell us about a chat exchange, show pictures of the involved, disclose nicknames from the phone book and so on.

These digital evidence represent a valuable source of analysis but… they can be altered any time or not fully recovered.

CDR have the highest level of inalterability compared with other sources

MULTILEVEL CORRELATION OF THE DIGITAL EVIDENCE

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Key list to understand CDR investigation

- ✔ Carriers
- ✔ Data retention
- ✔ IMSI
- ✔ IMEI
- ✔ CGI
- ✔ Cell tower coverage
- ✔ Cell towers lists
- ✔ User CDR
- ✔ Subscriber CDR
- ✔ Traffic CDR
- ✔ IMEI/IMSI
- ✔ Events type
- ✔ CDR file format
- ✔ CDR enquiry
- ✔ Integrity of the data
- ✔ HASH code
- ✔ Mobile extractions
- ✔ GPS tracks
- ✔ ...

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Who is providing the service?

Let’s talk about carriers

The cellular service provider covers with its network infrastructure the country, providing wireless communications service up to the latest technologies available.

What is:

- ✔ **MNO** (Mobile Network Operator)
- ✔ **MVNO** (Mobile Virtual Network Operator)



How long are CDR stored (available)?

DATA RETENTION refers to the records storage and preservation time, according to the country regulations.

Carriers collect CDR in their infrastructures. Each one, within limits provided by law, maintains the records for a defined period of time.



For example:

- ✔ Vodafone IT (the current legislation moved up to 6 years...)
- ✔ Verizon Wireless US (call history 7 years, tower dumps 1 year

General definition – CDR classification

Different type of CDR:

- ✔ User
- ✔ Traffic
- ✔ Cell towers lists
- ✔ IMEI/IMSI
- ✔ Subscribers
- ✔ …

Please note:

- ✔ Ask for any instruction sheet or legends to interpret the data
- ✔ Be aware of the period required and cell towers at the time of the facts

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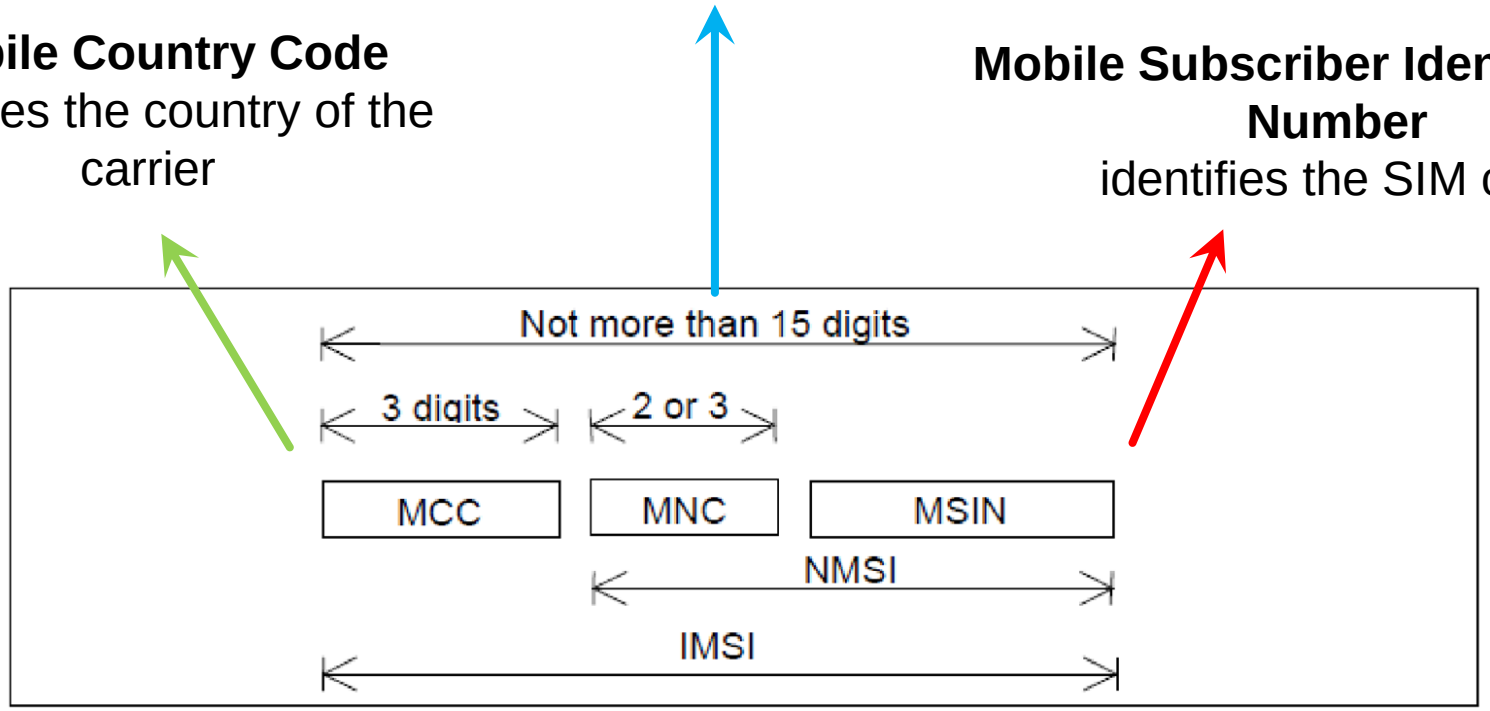
How can you recognize a device?

IMSI - International Mobile Subscriber Identity

Mobile Network Code
identifies the telephone carrier

Mobile Country Code
identifies the country of the carrier

Mobile Subscriber Identification Number
identifies the SIM card



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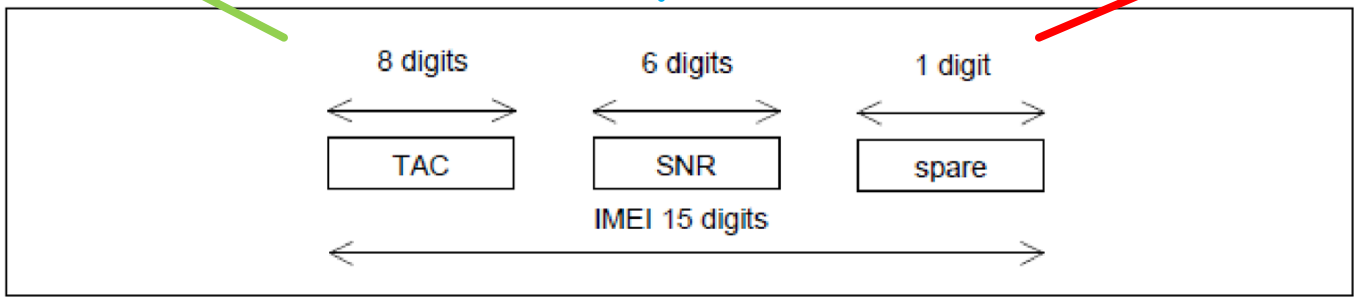
IMEI - International Mobile Station Equipment Identity

SN (Serial Number)
identifies the serial of the mobile

TAC (type allocation code)
it uniquely identifies manufacturer/model

CD (Check Digit)
verification code

***FAC (Final Assembly Code)**
Place of production/assembled (TAC+2 digits)



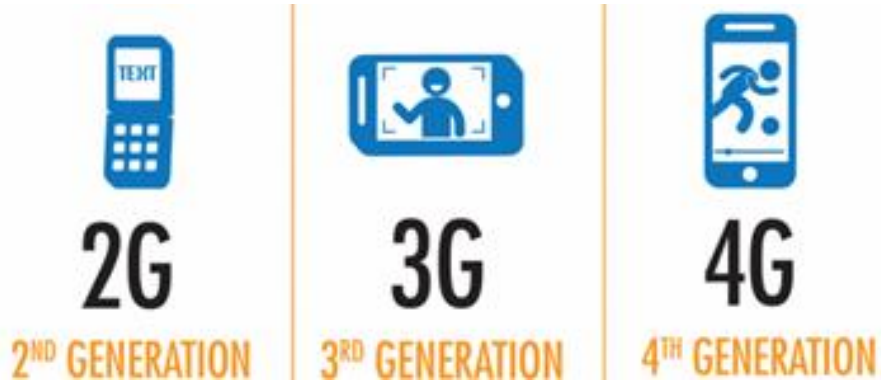
IMEISV - SVN (2 digits) = Software Version Number (identifies the preinstalled software version)

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Which is the infrastructure behind your service?

The Telco standards

- **2G** (GSM – global system for mobile communication)
- **3G** (UMTS – universal mobile telecommunication system)
- **4G** (LTE – long term evolution)
- Coming soon… 5G (fifth generation)



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Cell towers (BTS)

Base Transceiver Station:

- Mast/Pole
- Electronic equipment (antennas)



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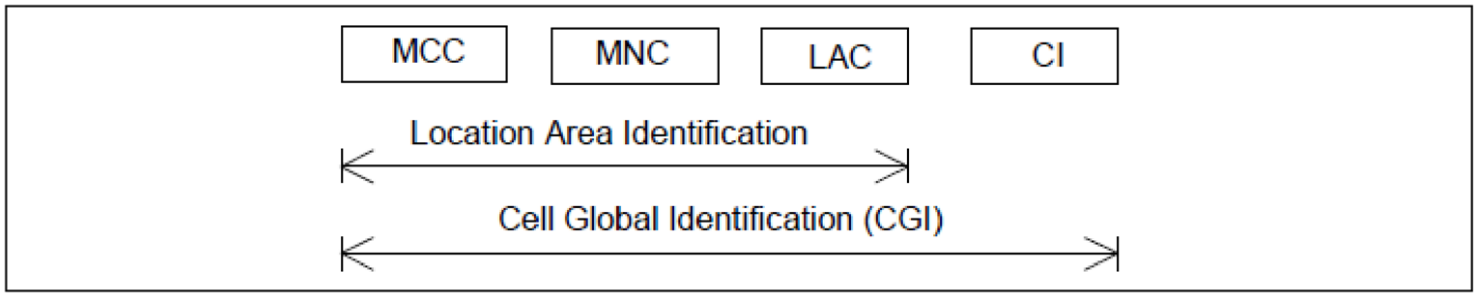
CGI - Cell Global Identity

Mobile Network Code
identifies the carrier

Location Area Code
identifies the area of the cell

Mobile Country Code
identifies the country of the carrier

Cell Identity
identifies the cell



Vodafone IT - LTE 222-10-37273-(000)13

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Cell Towers coverage
Sector and mapping

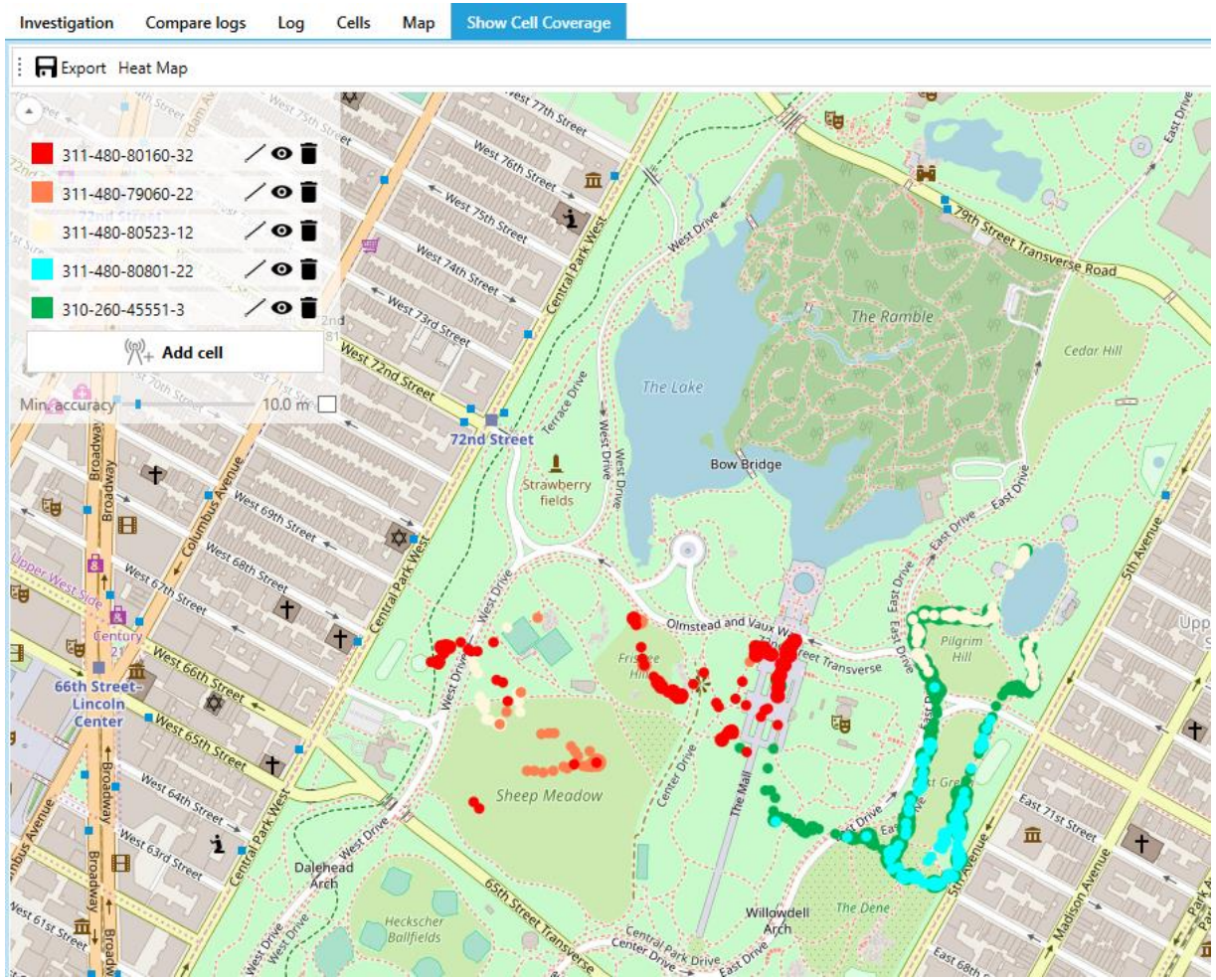
- Omni-directional > 360°
- Sectorized, a “sector” is specific portion of a geographic area covered by the cell
- Azimuth
- Beamwidth





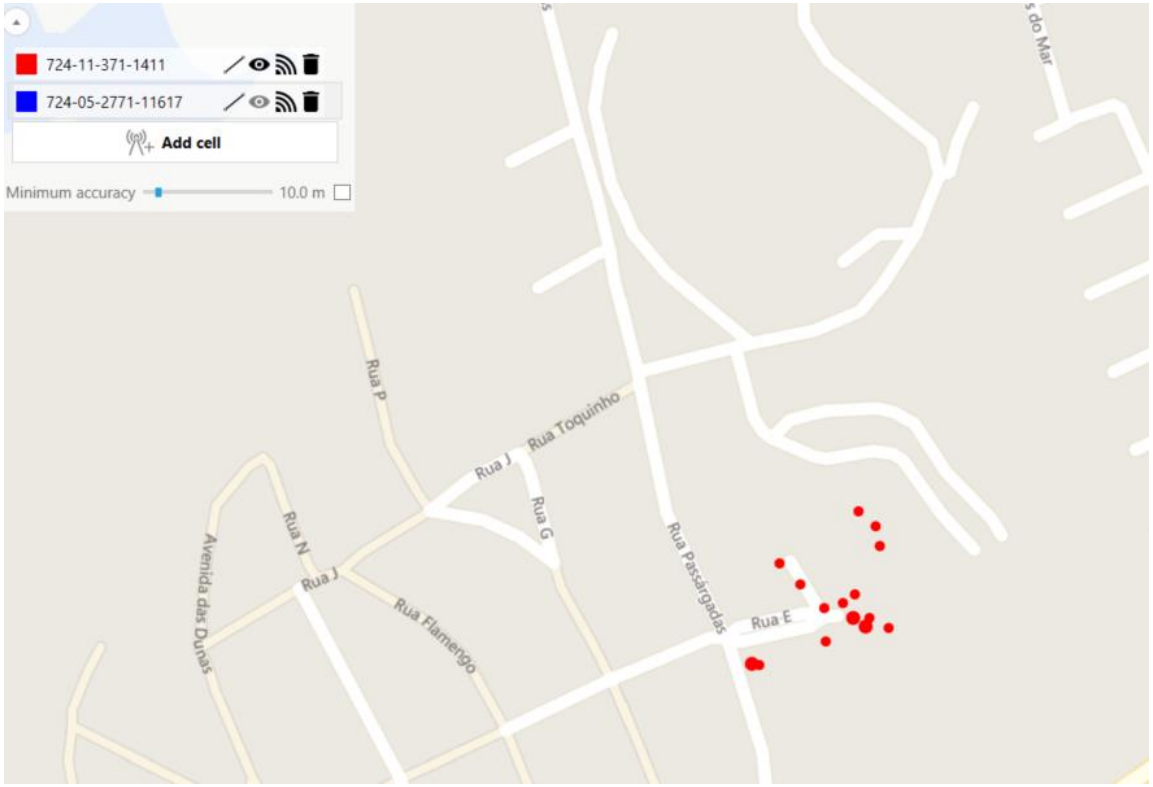
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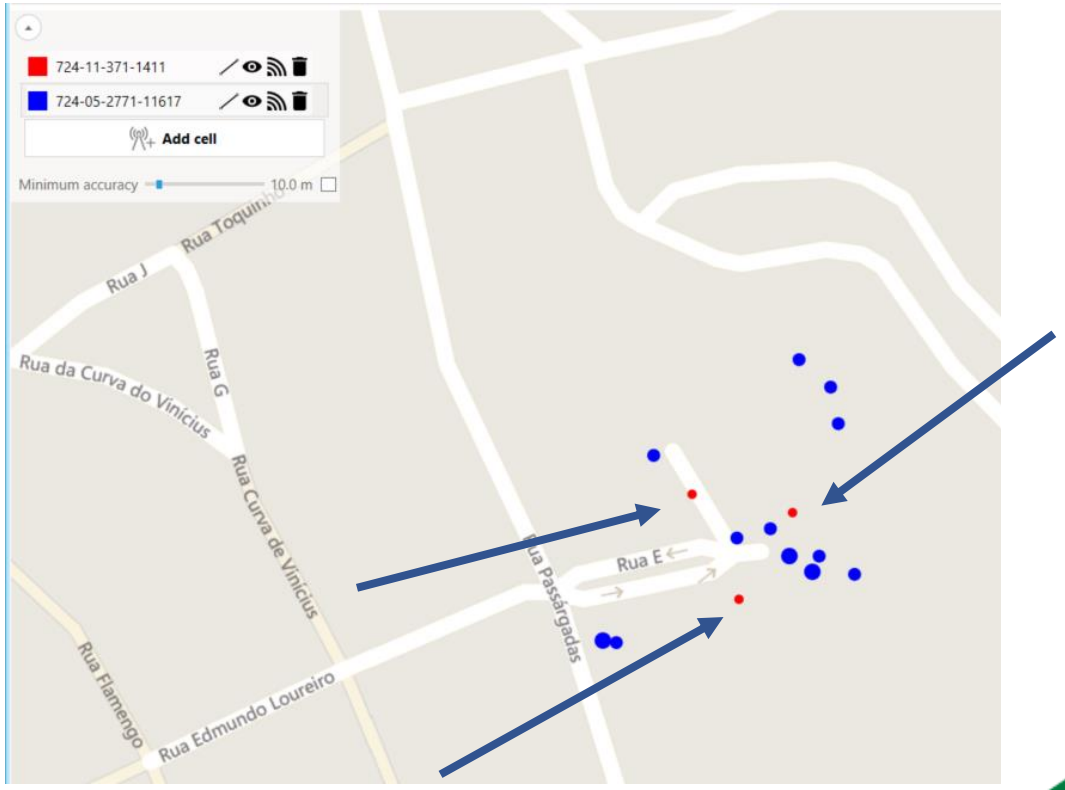
To which BTS are you connected?



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To which cell are you connected?





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Phone records from the carries

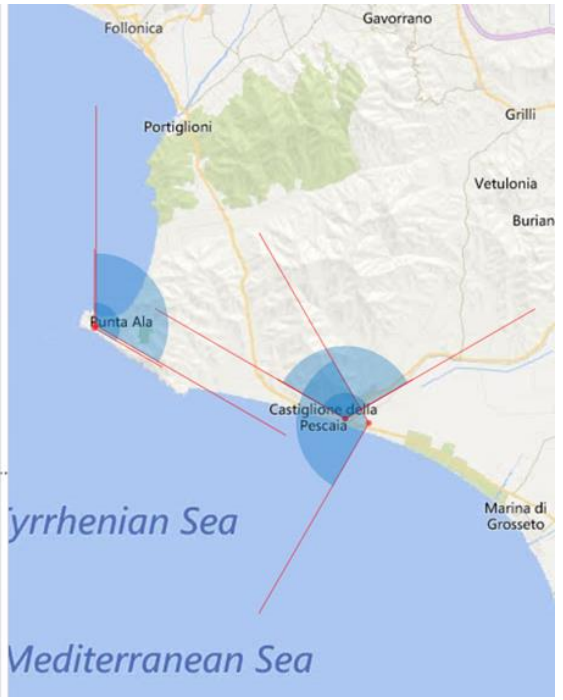
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Cell Towers list

It provides information about cell towers location and coverage area (latitude, longitude, address, city, azimuth etc.)

This is an example of Cell List imported and displayed using SecurCube®PhoneLog.

Code	Address	Village	District	Country	Latitude	Longitude	Network Ope	Technology	Azimuth
222-10-25041-13	Via Respighi > Via	MODENA	MO				DEMO1		
30025 - 4613	30025 - 4613	30025 - 4613	30025 - 4613				DEMO1		
222-10-35042-56	strada vicinale anti	CASTIGLIONE DELI	GR		42.76167	10.8922	DEMO2	UMTS	270
222-10-30096-36	strada vicinale anti	CASTIGLIONE DELI	GR		42.76267	10.89314	DEMO2		270
222-10-35042-61	Loc. Poggio del Ba	CASTIGLIONE DELI	GR		42.80144	10.74025	DEMO2	UMTS	340
222-10-30096-36	Loc. Poggio del Ba	CASTIGLIONE DELI	GR		42.80244	10.74119	DEMO2	GSM	60
222-10-35042-61	Loc. Poggio del Ba	CASTIGLIONE DELI	GR		42.80144	10.74025	DEMO2	UMTS	60
222-10-35042-56	Via Giatti, 8	CASTIGLIONE DELI	GR		42.76447	10.88001	DEMO2	UMTS	0
222-10-31061-42	Piazza Castello 11	SASSARI	SS		40.72621	8.562088	DEMO2	GSM	180
222-10-31061-47	Via Buddi Buddi, 4	SASSARI	SS		40.74789	8.557333	DEMO2	GSM	30
222-10-31061-98	Strada 19 Zona Inc	SASSARI	SS		40.734	8.53987	DEMO2		60
222-10-35042-39	Stadio comunale d	SANTA FIORA	GR		42.83	11.58	DEMO2	UMTS	270
222-10-30096-36	Via Giatti, 8	CASTIGLIONE DELI	GR		42.76447	10.88001	DEMO2	GSM	0
222-10-35032-56	Torrefaro nel parch	PISA	PI		43.70123	10.40123	DEMO2	UMTS	120
222-10-30096-70	S.S. 322 Delle Colla	SCARLINO	GR		42.85036	10.81289	DEMO2	GSM	155
222-10-35032-56	Torrefaro nel parch	PISA	PI		43.70123	10.40123	DEMO2	UMTS	0
222-10-35042-10	Cimitero comunale	CASTELLINA MARI	PI		43.4173	10.57036	DEMO2	UMTS	240
222-10-35032-56	V.le Gramsci, 14	PISA	PI		43.71053	10.40162	DEMO2	UMTS	240
222-10-35042-56	Loc. " Il Fico" - Cor	FOLLONICA	GR		42.93694	10.74861	DEMO2	UMTS	110
222-10-35042-23	VIA MARCONI (PR	GAVORRANO	GR		42.94	10.88	DEMO2	UMTS	160
222-10-35052-50	Loc. Sa Segada	ALGHERO	SS		40.63242	8.301567	DEMO2	UMTS	90
222-10-30096-71	C/o il cinema estiv	CASTIGLIONE DELI	GR		42.76278	10.87944	DEMO2	GSM	180
222-10-35042-56	Poggio Staffo	CASTIGLIONE DELI	GR		42.78205	10.8347	DEMO2	UMTS	350
222-10-30096-47	Via Adamello 22	GROSSETO	GR		42.77056	11.11267	DEMO2	GSM	60



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Cell Towers list - Example

2	MSC	BSC	CGI	Location Numb	CELLA	NOME SITO	Indirizzo	Comune	Provincia	Regione	Latitudine WGS84	Longitudine WGS84	Dir Irr	Altezza SLM	Altezza Antenna	Tilt	Tilt Elettrico Reale
3	USRVAN001	GBSCAN005	222-01-25286-62553	4-3971062553	AN22S	ANCONA (COLLE AMENO)	LOCALITA COLLE AMENO,S.N.C.	ANCONA	ANCONA	MARCHE	43 36 33	13 26 36.	ND	53			
4	USRVAN001	GBSCAN005	222-01-25286-62554	4-3971062554	AN22S	ANCONA (COLLE AMENO)	LOCALITA COLLE AMENO,S.N.C.	ANCONA	ANCONA	MARCHE	43 36 33	13 26 36.	ND	53			
5	USRVAN001	GBSCAN010	222-01-25285-62653		AN47S	SENIGALLIA SUD MARE	LUNGOMARE DANTE ALIGHIERI,98	SENIGALLIA	ANCONA	MARCHE	ND	ND	ND	3			
6	USRVAN001	GBSCAN010	222-01-25285-62654		AN47S	SENIGALLIA SUD MARE	LUNGOMARE DANTE ALIGHIERI,98	SENIGALLIA	ANCONA	MARCHE	ND	ND	ND	3			
7	USRVAN001	GBSCAN010	222-01-25285-62655		AN47S	SENIGALLIA SUD MARE	LUNGOMARE DANTE ALIGHIERI,98	SENIGALLIA	ANCONA	MARCHE	ND	ND	ND	3			
8	USRVAN001	GBSCAN011	222-01-25285-63661		PS43S1	URBINO TELECOM	VIA CALAMANDREI,S.N.C.	URBINO	PESARO E URBINO	MARCHE	ND	ND	ND	371			
9	USRVAN001	GBSCAN011	222-01-25285-63662		PS43S2	URBINO TELECOM	VIA CALAMANDREI,S.N.C.	URBINO	PESARO E URBINO	MARCHE	ND	ND	ND	371			
10	USRVAN001	GBSCAN013	222-01-25286-00088	4-39732000088	AC24D	CANCELLI DI FABRIANO FS	C/O PALO RFI STAZIONE CANCELLI DI FABRIANO,S.N.C	FABRIANO	ANCONA	MARCHE	43 18 03	12 49 54.	62	458	27,8	0,70	
11	USRVAN001	GBSCAN013	222-01-25286-00089	4-39732000089	AC24D	CANCELLI DI FABRIANO FS	C/O PALO RFI STAZIONE CANCELLI DI FABRIANO,S.N.C	FABRIANO	ANCONA	MARCHE	43 18 03	12 49 54.	260	458	27,9	1,50	
12	USRVAN001	GBSCAN013	222-01-25286-34593	4-39732034593	AN72D	CANCELLI	LOCALITA FOGLIO N.127 PARTICELLA 329,S.N.C.	FABRIANO	ANCONA	MARCHE	43 18 24	12 49 52.	3	556	10,3	2,50	
							LOCALITA FOGLIO N.127										

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User CDR – key elements

- ✔ **Who**, namely the caller/sender
- ✔ **What**, type of event
- ✔ **When**, information about date/duration and events
- ✔ **Which**, device and network operator, IMEI+IMSI
- ✔ **Where**, information given by the Cell Global Identity (CGI)
- ✔ **To Who**, namely the callee/called/receiver
- ✔ …



User CDR – Example

vodafone

RICERCA TRAFFICO STORICO

Traffico Telefonico

Pagina 1 di

Utenza: 347

Periodo: dal 21/02/2011 00:00:00 al 15/06/2012 06:30:00 **Ordinamento:** Data e ora

Prefissi internazionali: 0039 39

Prefissi locali:

Origine / Smcs	Chiamato	Chiamante	Data	Data Fine	Tipo	Causale	IMEI/SERIAL	IMSI	LAI-CI, Zona-Cella
	347 8	3357 7	21/02/2011 09:21:05	21/02/2011 09:21:28	2		353382 0	222107 13	3504 CAST - Loc. Sett.
	347 8	3351 0	21/02/2011 09:23:29	21/02/2011 09:33:26	2		353382 0	222107 13	3504 CAST - Loc. Sett.
	347 8	3351 0	21/02/2011 09:23:30	21/02/2011 09:33:26	1		354217 0	222105 47	3500 Franc Sett.
	347 8	3351 0	21/02/2011 09:38:08	21/02/2011 09:38:44	2		353382 0	222107 13	3504 CAST - Loc. Sett.
	347 8	3351 0	21/02/2011 09:38:09	21/02/2011 09:38:43	1		354217 0	222105 47	3500 Franc Sett.
	347 3	3473 8	21/02/2011 10:12:55	21/02/2011 10:14:09	1		353382 0	222107 13	3504 CAST - Loc. Sett.
	347 8	3476 3	21/02/2011 10:15:27	21/02/2011 10:16:43	2		353382 0	222107 13	3504 CAST - Loc. Sett.
	057	3473 8	21/02/2011 10:17:54	21/02/2011 10:22:09	1		353382 0	222107 13	3504 CAST Loc. Sett.
	340 3	3473 8	21/02/2011 10:23:07	21/02/2011 10:24:21	2		354838 0	222103 12	3007 A. Ma

Legenda: 1 Uscente voce; 2 Entrante voce; 3 Call forward; 4 Uscente voce estero; 5 Entrante voce estero; 6 Uscente sms; 7 Entrante sms; 8 Uscente sms estero; 9 Entrante sms estero; A Uscente v call-conference; B Entrante voce call-conference; C Uscente dati; D Entrante dati; E Uscente fax; F Entrante fax; G Call forward dati; H Call forward fax; I Uscente dati estero; J Entrante dati estero; K Uscente fax estero; L Entrante fax estero; M Callback; N Gprs; Q gprs estero; R Uscente mms; T Uscente Video; U Entrante Video; V Call forward Video; W Wlan; X Entrante mms; Y Uscente Video Est Z Entrante Video Estero; A18 Uscente MMS Web; A19 Entrante MMS Web; A20 Uscente SMS Web; A21 Entrante SMS Web; U1 Libero; U2 Occupato; U3 Non Raggiungibile/Altro

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Subscribers CDR – key elements

- Phone number
- IMSI
- Activation/Deactivation date*
- Dealer

Subscriber’s details:

- ✔ Name/Surname
- ✔ Date/Place of birth
- ✔ Residence
- ✔ Personal
- ✔ ...



Subscriber CDR – Example

File Edit Format View Help

NUM.TEL SN	NUMERO SIM	Tipo	SS	Attivaz. Sim		Disattiv. Sim	Super	Interno	Mobile	DEALER	EVENTO	COGNOME CUST.	NOME CUST.	RESIDENZA CUST.	L/D	NASCITA CUST.	C.F.,
3351827497	d	8939107900048222226		A	d	24-10-2009	15-06-2012		N	316.0972 - Dealer1 SRL - Main Street 64 - 58100 - City			Bat Man				ZIP
3351976294	d	8939107900048225142		A	d	24-10-2009	15-06-2012		N	316.0972 - Dealer1 SRL - Main Street 64 - 58100 - City			Ro Bin			ZIP code1 -	
3407265791	d	8939107900043635577		A	d	24-10-2009	15-06-2012		N	316.9N72 - Dealer4 SRL - NorthPole - 58100 - City4			Riutilizzo del numero			Santa Claus	
3357651591	d	8939107900048225121		A	d	24-10-2009	15-06-2012		N	316.0972 - Dealer1 SRL - Main Street 64 - 58100 - City			Ro Bin			ZIP code1 -	
3357655591	d	8939107900048225111		A	d	24-10-2009	15-06-2012		N	316.0972 - Dealer1 SRL - Main Street 64 - 58100 - City			Bat Man			ZIP code1 -	
3454201695	d	8939104700028372416		A	d	05-08-2010	14-12-2011		N	307.0557 - Dealer2 SRL - 51st Street 22 - 56014 - City2			Super Man				ZIP
3476695196	d	8939105400011071215		R	d	21-02-2001	16-06-2011		N	208.0491 - Dealer2 SRL - 51st Street 22 - 59100 - City2			Sherlock Holmes				ZIP
3453726694	a	8939103100003194117		A	a	01-04-2010		N		336.5975 - Dealer3 SRL - WallStrett 1111 - 28024 - City3			Clark Kent				ZIP
3489005597	d	8939107800052031522		A	d	20-03-2008	13-12-2012		N	316.7F64 - Dealer5 SRL - Hawaii - City5			Usain Bolt			ZIP code5 - Jamaica	
3473825592	d	8939107800345678522		A	d	20-03-2008	13-12-2012		N	316.7F64 - www.securcube.net - Paderno del Grappa - Asolo			TV	SecurCube			

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Traffic CDR – key elements

- ✔ Files that contain mobile events happened in a specific range of date(s)/time
- ✔ Includes all those connected to the cells in the area, doing somethings (mobile events)
- ✔ General outcome of traffic with no specific target (user/IMEI/IMSI)



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Traffic CDR – Example

Ricerca : 73[redacted] CONSULTAZIONE TRAFFICO RADIOMOBILE Telecom Italia S.p.A.

DATA	ORA	Durata	NCR/Call ID	Conf ID	Telefono Chte	Telefono Chto	Telefono Deviante	Imsi Deviante	Imsi Chte	Imsi Chto	Imei Chte	Imei Chto	MSC	Opest	Naz	Cellid
07/02/2018	11:30:05	000000			[redacted]											R5
07/02/2018	11:30:33	000030			[redacted]											R5
07/02/2018	11:30:55	000040	23C34007F9		[redacted]											R5
07/02/2018	11:31:17	000279			[redacted]											R5
07/02/2018	11:31:21	000010	23D30607F9		[redacted]											R5
07/02/2018	11:31:34	000070			[redacted]											R5
07/02/2018	11:31:37	000000			[redacted]											R5

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IMEI/IMSI – key elements

- ✔ Specific target. Request done using a known device/SIM.
- ✔ It contains the same elements of a User CDR



Events type

The mobile events (call, SMS, data session etc.) in the CDR come in codes/digits. They can be red using a key line, in general, available with the CDR, given by carriers.

Origine / Smcs	Chiamato	Chiamante	Data	Data Fine	Tipo	Causale	IMEI/SERIAL	IMSI	LAI-CI, Zona-Cella
	347 8	3357 7	21/02/2011 09:21:05	21/02/2011 09:21:28	2		353382 0	222107 13	3504 CAST Loc. Sett.
	347 8	3351 0	21/02/2011 09:23:29	21/02/2011 09:33:26	2		353382 0	222107 13	3504 CAST Loc. Sett.
	347 8	3351 0	21/02/2011 09:23:30	21/02/2011 09:33:26	1		354217 0	222105 47	3500 Franc Sett.

CLARO	Llamadas Entrantes	
MOVISTAR	S LLamadas	
MOVISTAR	E LLamadas	
MOVISTAR	T LLamadas	
MOVISTAR	Mensaje sin cargo	
MOVISTAR	Mensajes Terminados con Origen en TP	
MOVISTAR	Serv. Mensajes de Texto	
MOVISTAR	MT Premium	
MOVISTAR	Mensajes Terminados con Origen en Mov	
MOVISTAR	Mensajes Gratis	

Events description using SecurCube®PhoneLog

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What about CDR formats?

- ✔ Plain text (txt)
- ✔ Csv
- ✔ Excel
- ✔ PDF
- ✔ Rft
- ✔ Doc
- ✔ …

In spite of the fact that the files have similar basic information (date, time, duration etc.), the format changes… daily too.

Plain text

- ✔ Different NewLine for each OS
- ✔ No conventions about data placement

```
************************************************
H3G S.p.A.
************************************************

************************************************
Anagrafica per  MSISDN
************************************************
Msisdn = +3939
Data Iniziale = 03/03/2003 00:00:00
Data Finale = 30/06/2010 23:59:59

************************************************
Stato Anagrafico precedente alla data inizio
************************************************
Nessuna informazione disponibile per il cliente

Nessuna informazione disponibile per la linea

************************************************
Variazione Anagrafica Cliente del 25/06/2009 19:56:59
************************************************
Dati Anagrafici:
CF/P-IVA:
Indirizzo:
Luogo-Data Nascita:
Documento: C.I Nr.:
Anno Costituzione:
Registro Imprese:

************************************************
Variazione Anagrafica Linea
************************************************
Utenza : +3939
    Stato Utenza: Activated
    Data Att.Utenza: 26/06/2009
    Data Disatt.Utenza:

Sim Card - IMSI: 22299
    ICCID: 893
    Stato IMSI: Activated
    Data Ass.Anag.IMSI: 26/06/2009
    Data Disatt.IMSI:
```

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CSV

- ✔ Comma-Separated Values
- ✔ One row/record = one text row
- ✔ Double quotes to enclose strings with escapes
- ✔ Different separation values



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Excel

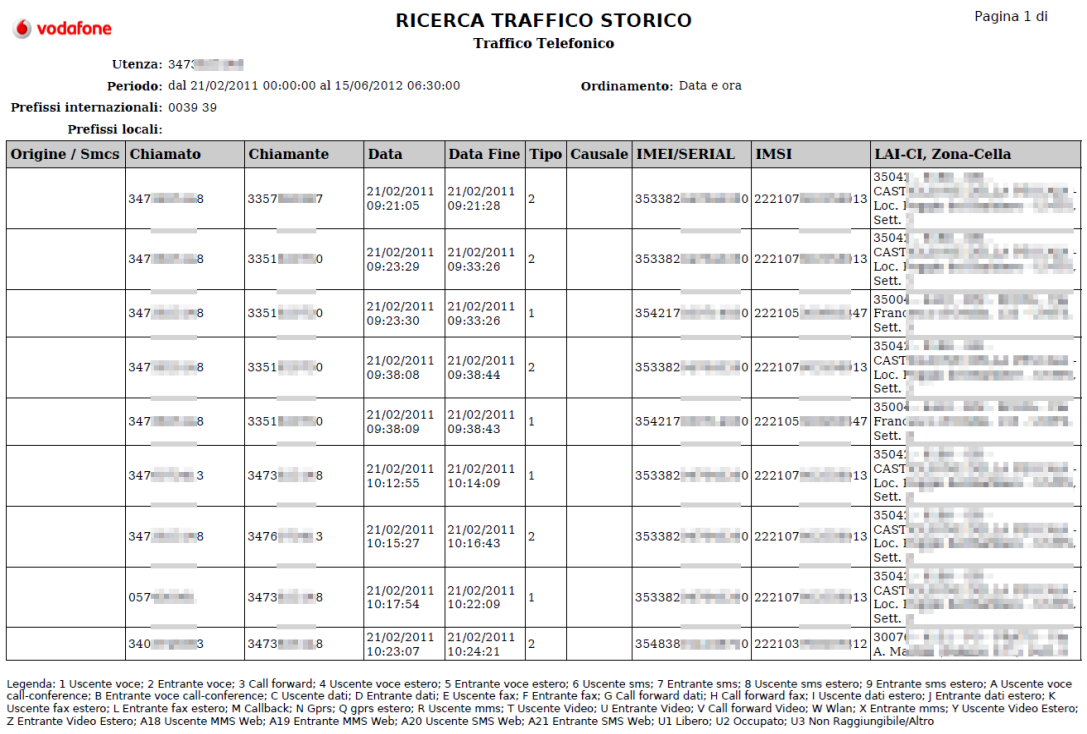
- ✔ SpreadSheet
- ✔ Direct correspondence with CSV

2	MSC	BSC	CGI	Location Numb	CELLA	NOME SITO	Indirizzo	Comune	Provincia	Regione	Latitudine WGS84	Longitudine WGS84	Dir Irr	Altezza SLM	Altezza Antenna	Tilt	Tilt Elettrico Reale
3	USRVAN001	GBSCAN005	222-01-25286-62553	4-3971062553	AN22S	ANCONA (COLLE AMENO)	LOCALITA COLLE AMENO,S.N.C.	ANCONA	ANCONA	MARCHE	43 36 33	13 26 36.	ND	53			
4	USRVAN001	GBSCAN005	222-01-25286-62554	4-3971062554	AN22S	ANCONA (COLLE AMENO)	LOCALITA COLLE AMENO,S.N.C.	ANCONA	ANCONA	MARCHE	43 36 33	13 26 36.	ND	53			
5	USRVAN001	GBSCAN010	222-01-25285-62653		AN47S	SENIGALLIA SUD MARE	LUNGOMARE DANTE ALIGHIERI,98	SENIGALLIA	ANCONA	MARCHE	ND	ND	ND	3			
6	USRVAN001	GBSCAN010	222-01-25285-62654		AN47S	SENIGALLIA SUD MARE	LUNGOMARE DANTE ALIGHIERI,98	SENIGALLIA	ANCONA	MARCHE	ND	ND	ND	3			
7	USRVAN001	GBSCAN010	222-01-25285-62655		AN47S	SENIGALLIA SUD MARE	LUNGOMARE DANTE ALIGHIERI,98	SENIGALLIA	ANCONA	MARCHE	ND	ND	ND	3			
8	USRVAN001	GBSCAN011	222-01-25285-63661		PS43S1	URBINO TELECOM	VIA CALAMANDREI,S.N.C.	URBINO	PESARO E URBINO	MARCHE	ND	ND	ND	371			
9	USRVAN001	GBSCAN011	222-01-25285-63662		PS43S2	URBINO TELECOM	VIA CALAMANDREI,S.N.C.	URBINO	PESARO E URBINO	MARCHE	ND	ND	ND	371			
10	USRVAN001	GBSCAN013	222-01-25286-00088	4-39732000088	AC24D	CANCELLI DI FABRIANO FS	C/O PALO RFI STAZIONE CANCELLI DI FABRIANO,S.N.C	FABRIANO	ANCONA	MARCHE	43 18 03	12 49 54.	62	458	27,8	0,70	
11	USRVAN001	GBSCAN013	222-01-25286-00089	4-39732000089	AC24D	CANCELLI DI FABRIANO FS	C/O PALO RFI STAZIONE CANCELLI DI FABRIANO,S.N.C	FABRIANO	ANCONA	MARCHE	43 18 03	12 49 54.	260	458	27,9	1,50	
12	USRVAN001	GBSCAN013	222-01-25286-34593	4-39732034593	AN72D	CANCELLI	LOCALITA FOGLIO N.127 PARTICELLA 329,S.N.C.	FABRIANO	ANCONA	MARCHE	43 18 24	12 49 52.	3	556	10,3	2,50	
							LOCALITA FOGLIO N.127										

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PDF

- ✔ Portable Document Format
- ✔ Fixed formatting
- ✔ Difficult to read for non-humans



vodafone

RICERCA TRAFFICO STORICO

Traffico Telefonico

Pagina 1 di

Utenza: 347

Periodo: dal 21/02/2011 00:00:00 al 15/06/2012 06:30:00 **Ordinamento:** Data e ora

Prefissi internazionali: 0039 39

Prefissi locali:

Origine / Smcs	Chiamato	Chiamante	Data	Data Fine	Tipo	Causale	IMEI/SERIAL	IMSI	LAI-CI, Zona-Cella
	347 8	3357 7	21/02/2011 09:21:05	21/02/2011 09:21:28	2		353382 0	222107 13	3504 CAST Loc. I Sett.
	347 8	3351 0	21/02/2011 09:23:29	21/02/2011 09:33:26	2		353382 0	222107 13	3504 CAST Loc. I Sett.
	347 8	3351 0	21/02/2011 09:23:30	21/02/2011 09:33:26	1		354217 0	222105 47	3500 Franc Sett.
	347 8	3351 0	21/02/2011 09:38:08	21/02/2011 09:38:44	2		353382 0	222107 13	3504 CAST Loc. I Sett.
	347 8	3351 0	21/02/2011 09:38:09	21/02/2011 09:38:43	1		354217 0	222105 47	3500 Franc Sett.
	347 3	3473 8	21/02/2011 10:12:55	21/02/2011 10:14:09	1		353382 0	222107 13	3504 CAST Loc. I Sett.
	347 8	3476 3	21/02/2011 10:15:27	21/02/2011 10:16:43	2		353382 0	222107 13	3504 CAST Loc. I Sett.
	057	3473 8	21/02/2011 10:17:54	21/02/2011 10:22:09	1		353382 0	222107 13	3504 CAST Loc. I Sett.
	340 3	3473 8	21/02/2011 10:23:07	21/02/2011 10:24:21	2		354838 0	222103 12	3007 A. Me

Legenda: 1 Uscente voce; 2 Entrante voce; 3 Call forward; 4 Uscente voce estero; 5 Entrante voce estero; 6 Uscente sms; 7 Entrante sms; 8 Uscente sms estero; 9 Entrante sms estero; A Uscente voce call-conference; B Entrante voce call-conference; C Uscente dati; D Entrante dati; E Uscente fax; F Entrante fax; G Call forward dati; H Call forward fax; I Uscente dati estero; J Entrante dati estero; K Uscente fax estero; L Entrante fax estero; M Callback; N Gprs; O gprs estero; R Uscente mms; T Uscente Video; U Entrante Video; V Call forward Video; W Wlan; X Entrante mms; Y Uscente Video Estero; Z Entrante Video Estero; A18 Uscente MMS Web; A19 Entrante MMS Web; A20 Uscente SMS Web; A21 Entrante SMS Web; U1 Libero; U2 Occupato; U3 Non Raggiungibile/Altro

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Good to know in CDR analysis

CDR enquiry

First of all:

JUDGE - WARRANT - LIST OF CDR OF INTEREST* (be precise)

Then:

WAITING TIME - WEB PORTAL / CARRIERS REPLY



*[In general, the request is free of charge]

SecurCube®

Integrity of the data

Analyse digital evidence with the assurance that the original files remain uncorrupted is required for a professional outcome.

Neither software automatic modification during the examination nor unreliable storage sources are admissible.

Data retention is very strict and once data go out of the allowed period cannot be retrieved for any reason.

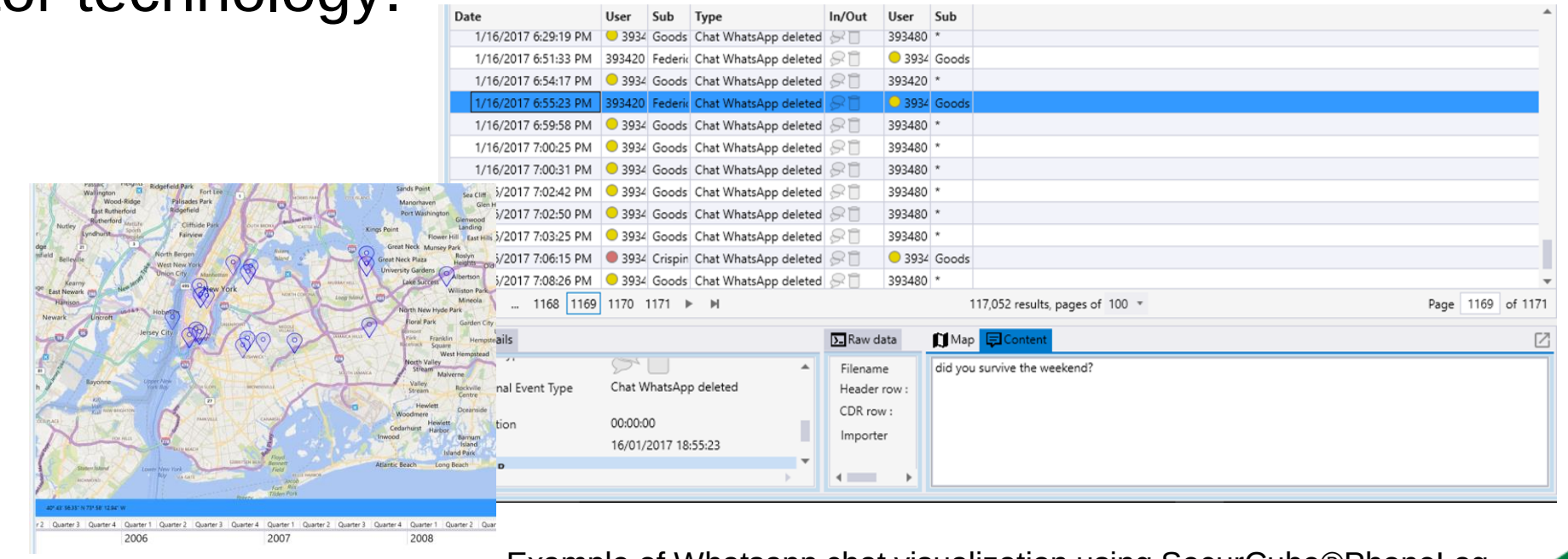


Mobile extraction

The content of a device (smartphone, tablet etc.) obtained using a dedicated extractor technology.

- contact list
- pictures
- chats
- SMS
- Email
- ...

To name a few.



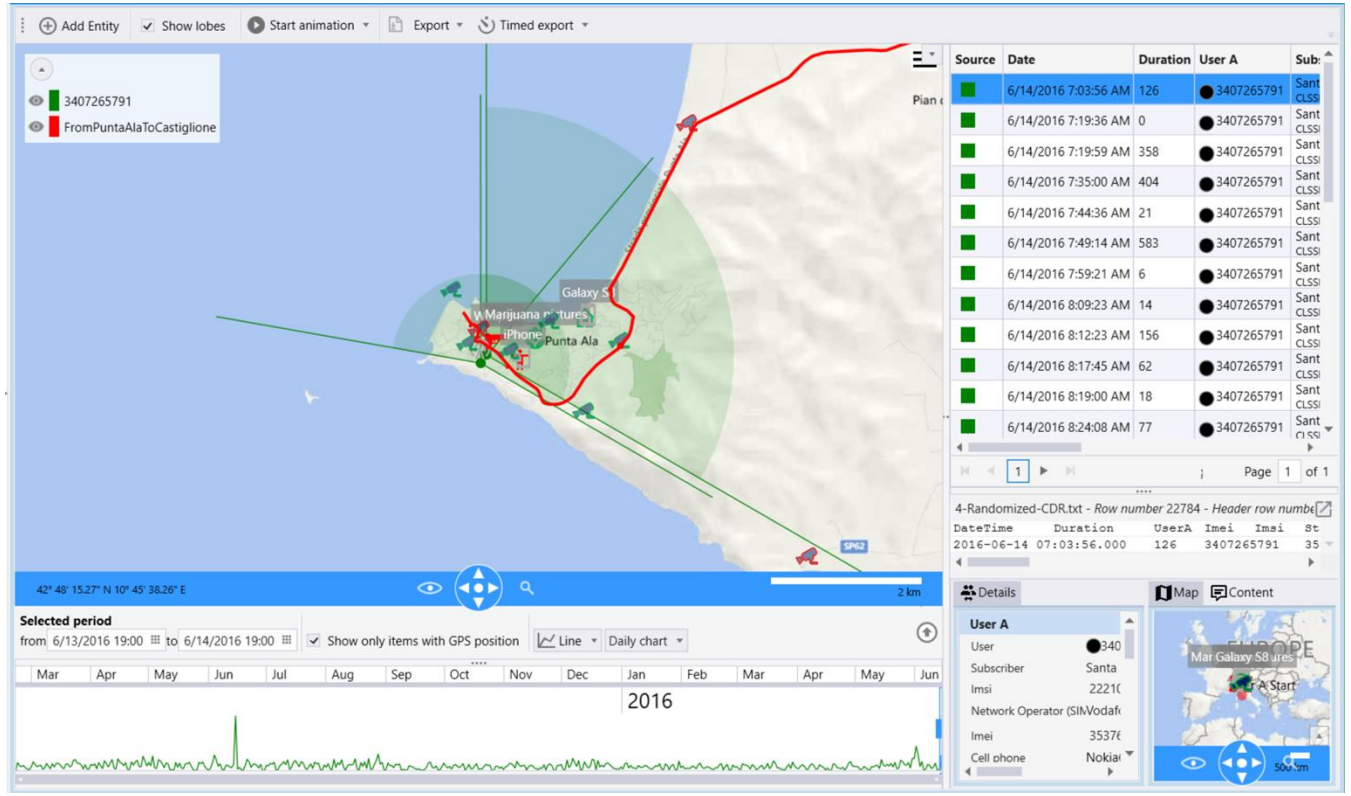
Example of Whatsapp chat visualization using SecurCube®PhoneLog

NOTE: in the majority of the cases, it is not possible to extract the 100% of the information from a device… ask for the CDR. It can assist you.

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GPS Tracks

Vehicles, Tracking devices etc. can tell you a story about places, habits, movements… Target daily routine.



Example of GSP and CDR correlation using SecurCube®PhoneLog

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The challenges in digital forensics examination

Some of the main threats:

- ✔ Anti-forensics considering that digital evidence could be subject to alteration
- ✔ Digital experts skills and tools in use
- ✔ Limited time to recover data (e.g. data retention). Huge amount of information need time, overnight procedures can compromise the investigation
- ✔ Legal aspects and regulation can change from country to country

- ✔ **The knowledge of cell site real coverage issues**

SecurCube®



Digital Forensic brand-new innovation

SecurCube Phone Log

Fast and Reliable CDRs(Call Detail Records) analysis

Digital Forensic
Brand New Innovation

The Digital Forensic is fast-growing and it is necessary to reply with efficient technologies and tools.

Mobile Device Analysis scenario

THE STARTING POINTS OF THE INVESTIGATION

What kind of data can you obtain from the device?

- Mobile extractions (images, videos, audios, messages etc.)
- CDRs (Call Detail Records – telephone operators)
- Cell coverage measurements

The purpose

To define the suspect’s profile discovering habits and movements. Confirm or refuse the alibi in relation to the crime event.

Digital Forensic brand-new innovation

SecurCube Phone Log

Fast and Reliable CDRs(Call Detail Records) analysis

Integration and correlation of the data with the powerful and Intuitive investigation software to make analysis of:

CDRs, Call Detail Records

+

Mobile extraction, UFED, OXYGEN Forensics, XRY

+

Log files of cell coverage measurements

WHY CHOOSE IT

Work easily. Always with the same tool.

Digital Forensic brand-new innovation

SecurCube Phone Log

Fast and Reliable CDRs(Call Detail Records) analysis

THE GOOD REASONS TO USE IT

Compatible

ALL AROUND THE WORLD CDRs and TELEPHONE OPERATORS

Reliable

DEVELOPED STARTING FROM DAILY LAW ENFORCEMENT INVESTIGATION EXPERIENCE

Flexible

INTEGRATION WITH DATA FROM OTHER SOURCES

Digital Forensic brand new innovation

SecurCube Phone Log

Fast and Reliable CDRs(Call Detail Records) analysis

THE SOFTWARE IS UP TO YOU

Easy to use

NO SPECIFIC TECHNICAL KNOWLEDGE REQUIRED TO RUN THE ANALYSIS

Interactive

RESULTS ON MAP VIEW AND REPORTS EXPORTABLE

Investigation grows with your insights

ADD NEW EVIDENCES GRADUALLY AND PROCEED WITH THE ANALYSIS

Let's talk about SecurCube Phone Log

What is a CDR - Call Detail Records file?

- ✔ File that summarizes all the mobile activity of the user (date/hours of inbound and outbound calls, SMS, connections, cell coverage and much more)
- ✔ Data collected with a row timeline logic
- ✔ Each record provides a specific result

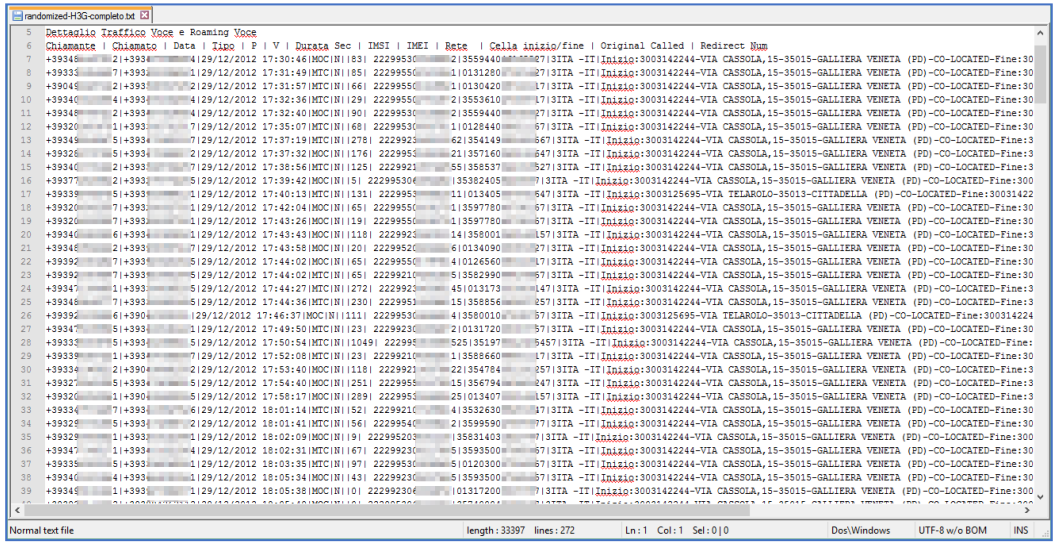


Fig. 1 Example of CDR file structure

How can the investigator analyze CDRs with Phone Log?

Step 1 - Obtain CDRs from the carriers

SecurCube Phone Log imports phone records natively in various formats with every kind of internal structure.

Step 2 - Make analysis (the digital investigation)

- Display users movements on the maps
- Evaluate phone activities among users
- Understand users' behaviors
- Find users with similar movement patterns

Step 3 - Generate reports

SecurCube Phone Log exports results in Csv, Excel, PDF, graphic format and also ZIP archives certified by HASH.

Importing CDRs fast Drag&Drop

SecurCube Phone Log file supported: TXT, CSV, XLS, XLSX, PDF

In addition to CDRs, the software can import:

- reports from UFED and OXYGEN forensics
- BTS Tracker logs
- GPS tracks

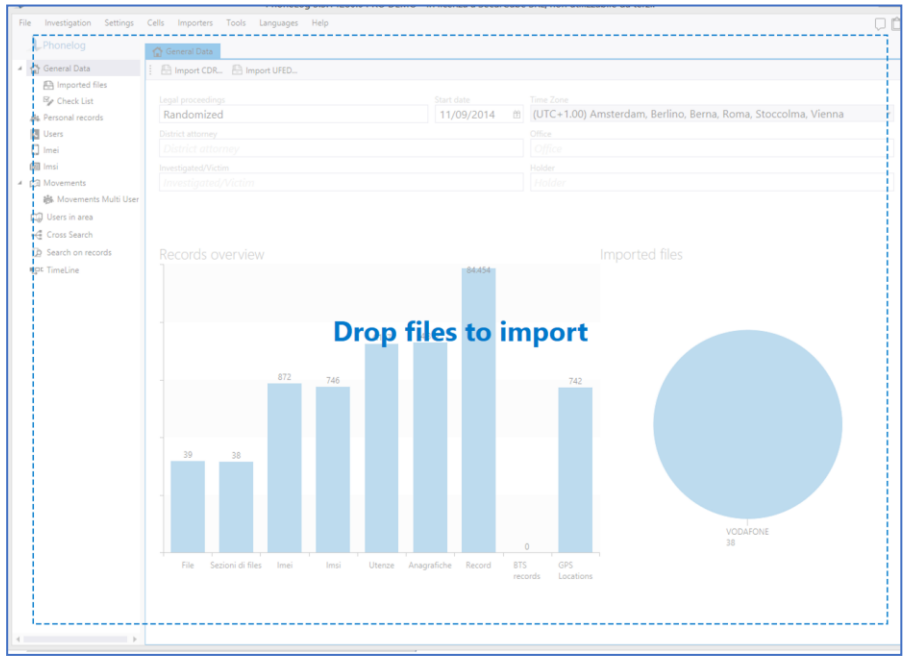


Fig. 3 Before to import the files

Strong Import/Parse capabilities

- ✔ SecurCube PhoneLog can import CDRs with any file structure
- ✔ Functionality “Custom CDR importer creation”
- ✔ The investigator can parse any kind of data log (also driving licence logs, highway traffic log and much more)
- ✔ New importers can be used to parse future CDRs

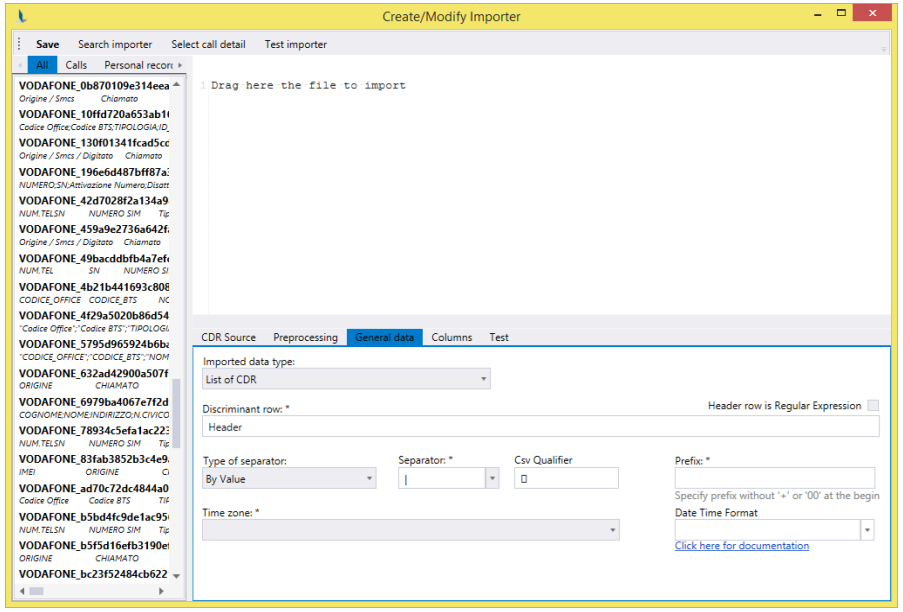


Fig. 4 Functionality “Custom CDR importer creation”

First Glance

This is the overview on the data imported.

Details about:

- Proceeding
- District
- Case holder
- Investigation start date

Correlation:

- time-zones
- CDRs from different countries
- operators

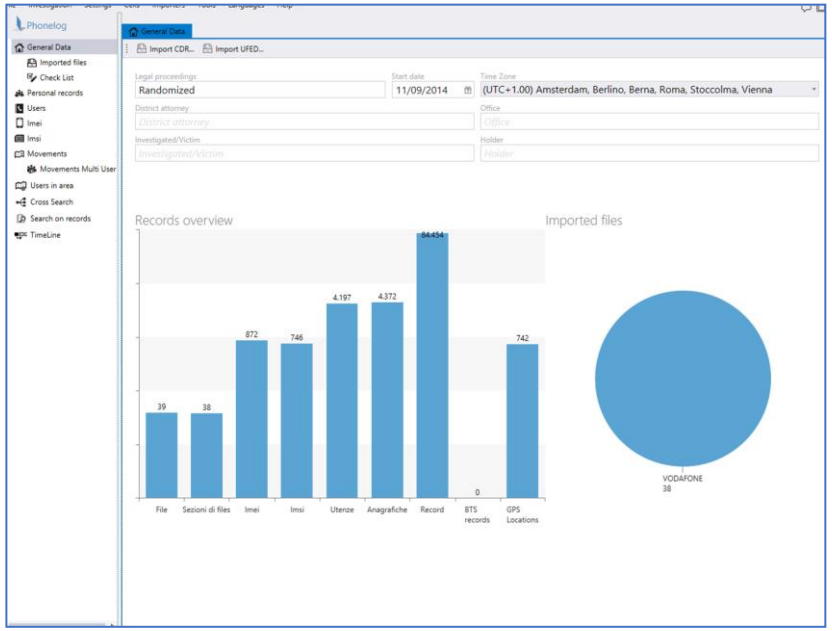


Fig. 5 General Data overview

Integration with Maps

- ✔ CDR files that have GPS position can be displayed by SecurCube Phone Log application in the map, with custom colors
- ✔ Ability to visualize different kinds of data in the map
- ✔ Detailed maps based on BingMaps
- ✔ High performance with on line and off line maps
- ✔ Suitable for other map providers (2GIS, Yandex)

Search on records [Standard research]

Clear search to slim down the list of CDR rows.

Cross the data to find correlation among:

- users
- telephone numbers
- cell coverage
- date and hours

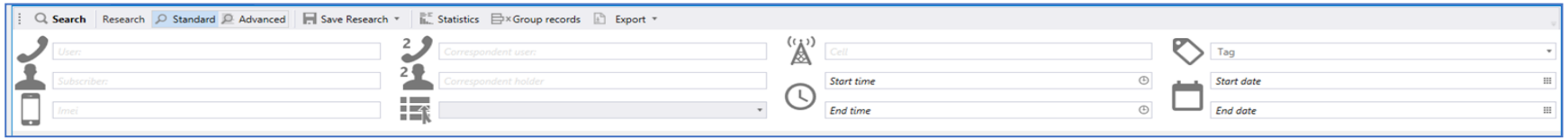


Fig. 6 Search on records criteria of search

Search on records [Standard research]

Right side area to summarize the results of the research (geolocation, phone information, correlations).

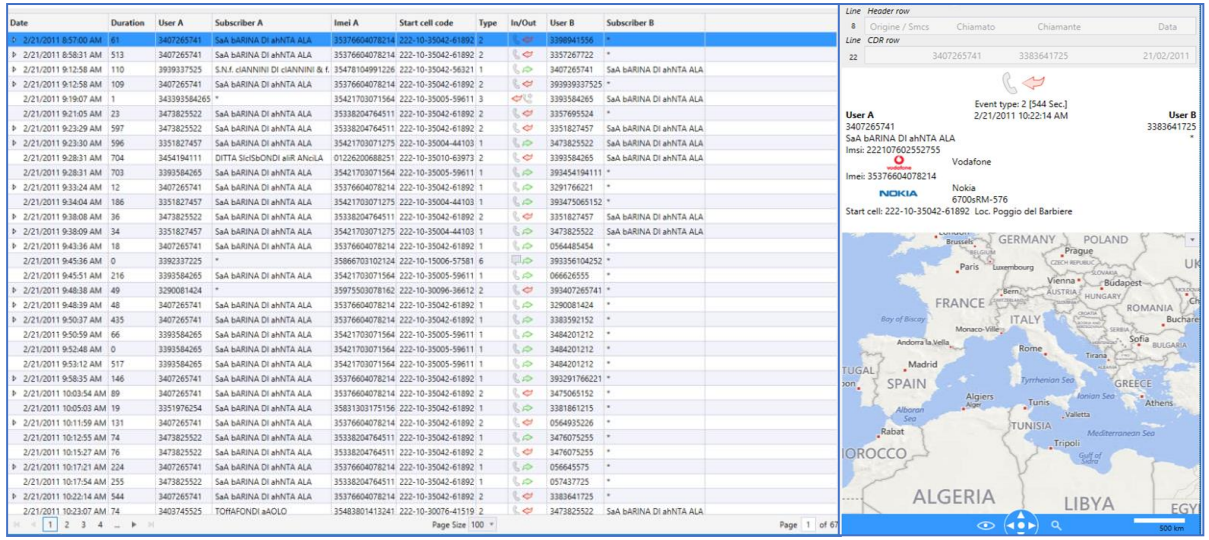


Fig. 7 List of records and map results

Search on records [Advanced research]

The investigator can compose his own search criterias (AND/OR):

- With any granularity
- With any tree depth
- Using several fields to filter by

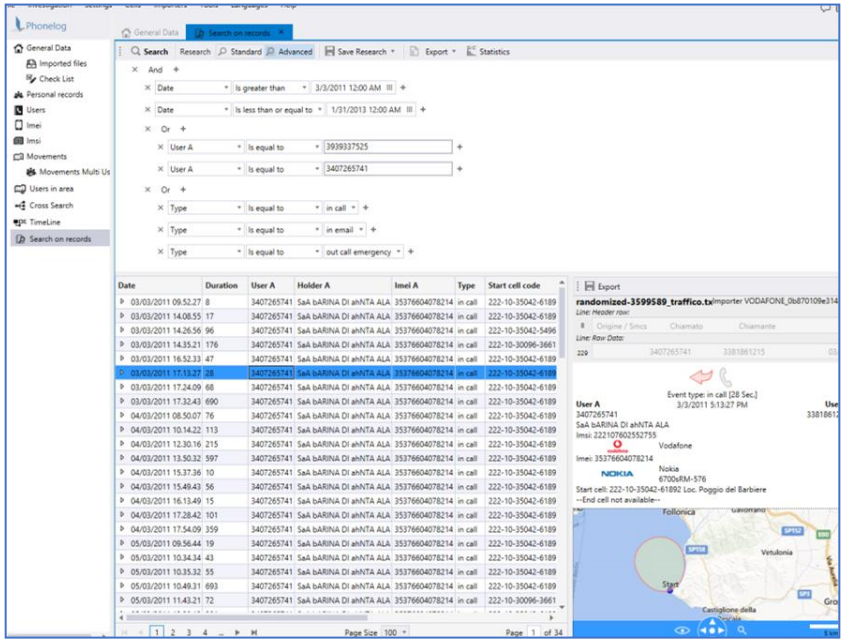


Fig. 8 Advanced search criterias

Users in area
When You know WHERE

When You know the area so “WHERE”, but not who/what.

The functionality:

- Draw areas and search what happened there
- Click on desired Records to make BTS blink

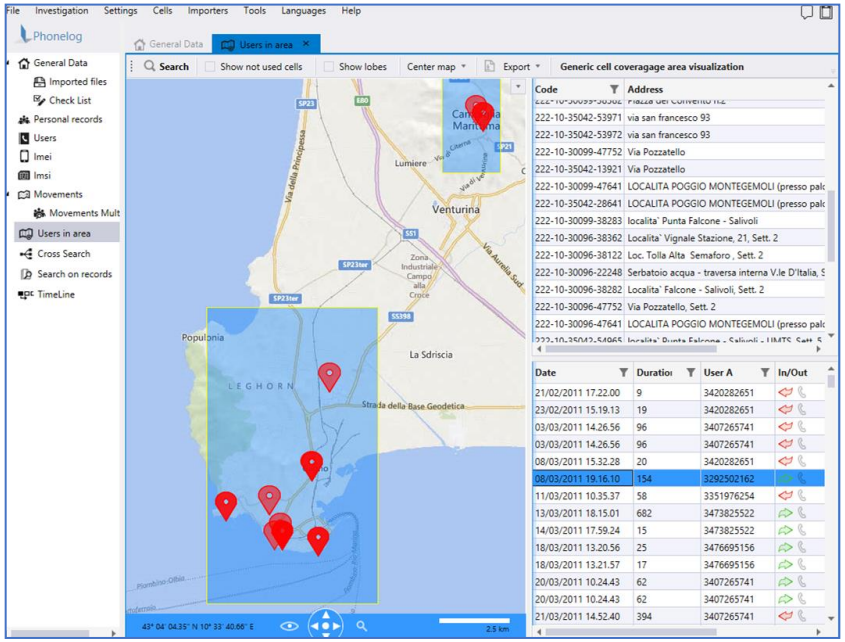


Fig. 9 Users in area functionality

Movements
When You know WHO

When You know the user, so “WHO”, but not Where.

The functionality:

- Track user’s movements
- Draw BTS on map following mouse movements over the horizontal calendar

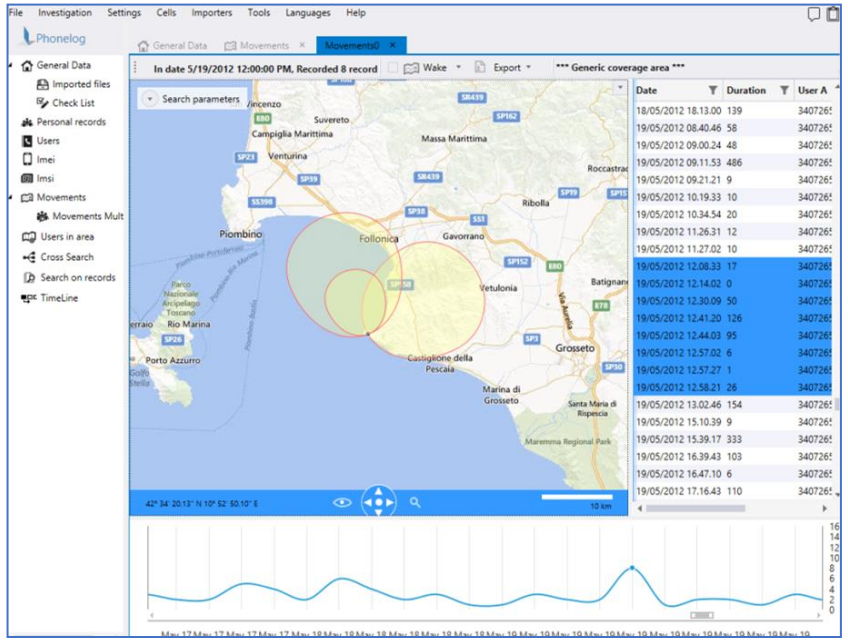


Fig. 10 Movements functionality

Movements Multi User
When You know [many] WHOs

The functionality tracks several user's movements at once

Different color for each user to distinguish in:

- Map
- Horizontal calendar
- Records list

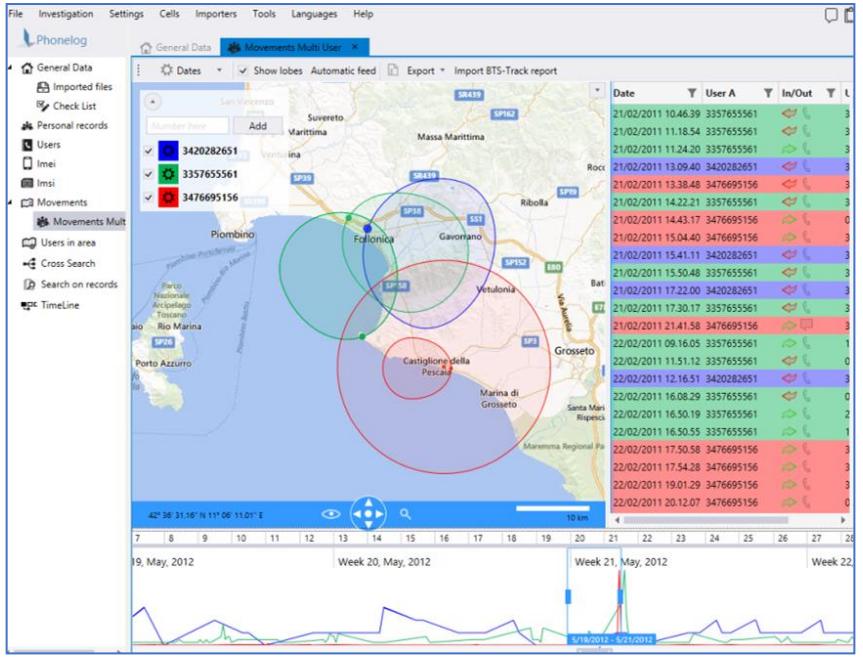


Fig. 11 Movements Multi User functionality

Cross Search data

Cross-check data is simple as do Drag&Drop. Powerful functionality to integrate in the reserach.

- What kind of correlation exist among RobberyAAA, TheftBBB and also from TheftCCC, in the CDRs?
- Automatic holder details evaluation for each result

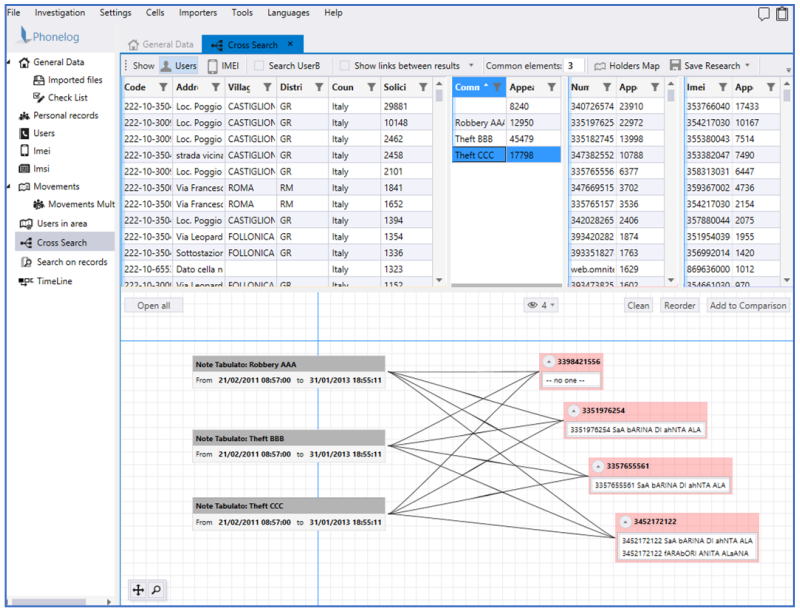


Fig. 12 Cross Search functionality

Cross Search heterogeneous data

Who ?

- made traffic under BTS 222-10-3009636611
- & made traffic under BTS 222-10-35042-56403
- & had contacts with User 3357655561
- & had contacts with IMEI 35195403948162

It evaluate results such as users or IMEI.

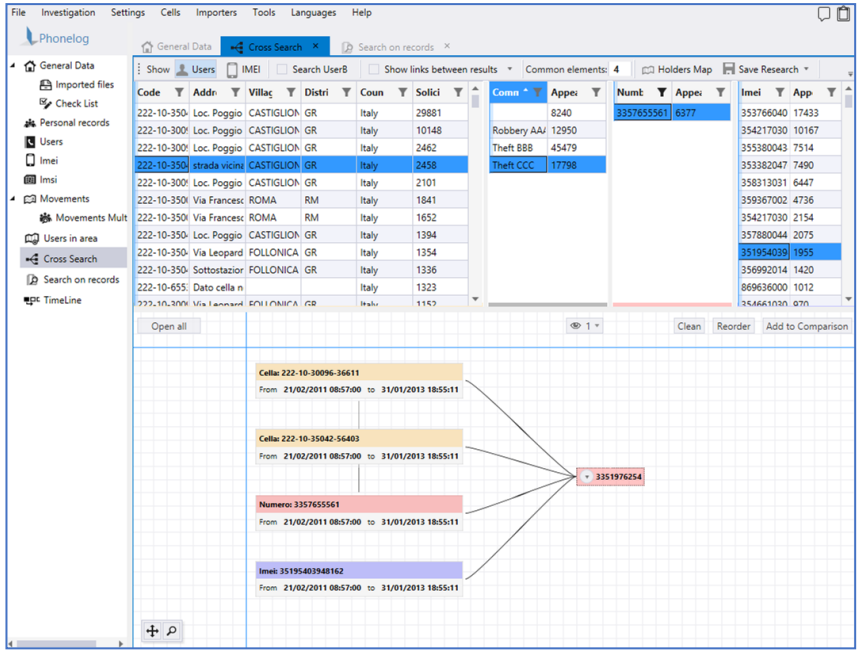


Fig. 13 Cross Search activity

Graphic Timeline

The functionality shows mobile activities with a timeline logic

Immediate comparison of the telephone activity made by:

- User
- IMEI
- IMSI

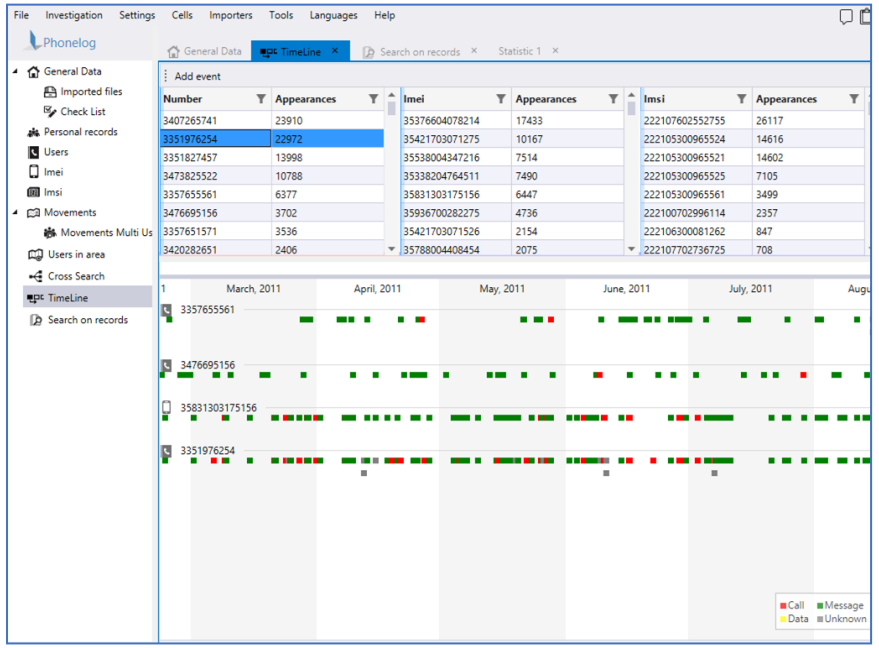


Fig. 14 Timeline overview

In depth:User's habits

To study specific user's habits and behaviors:

- Contacted numbers
- Peak of telephone activity
- What kind of activity
- IMEI IMSI contacted

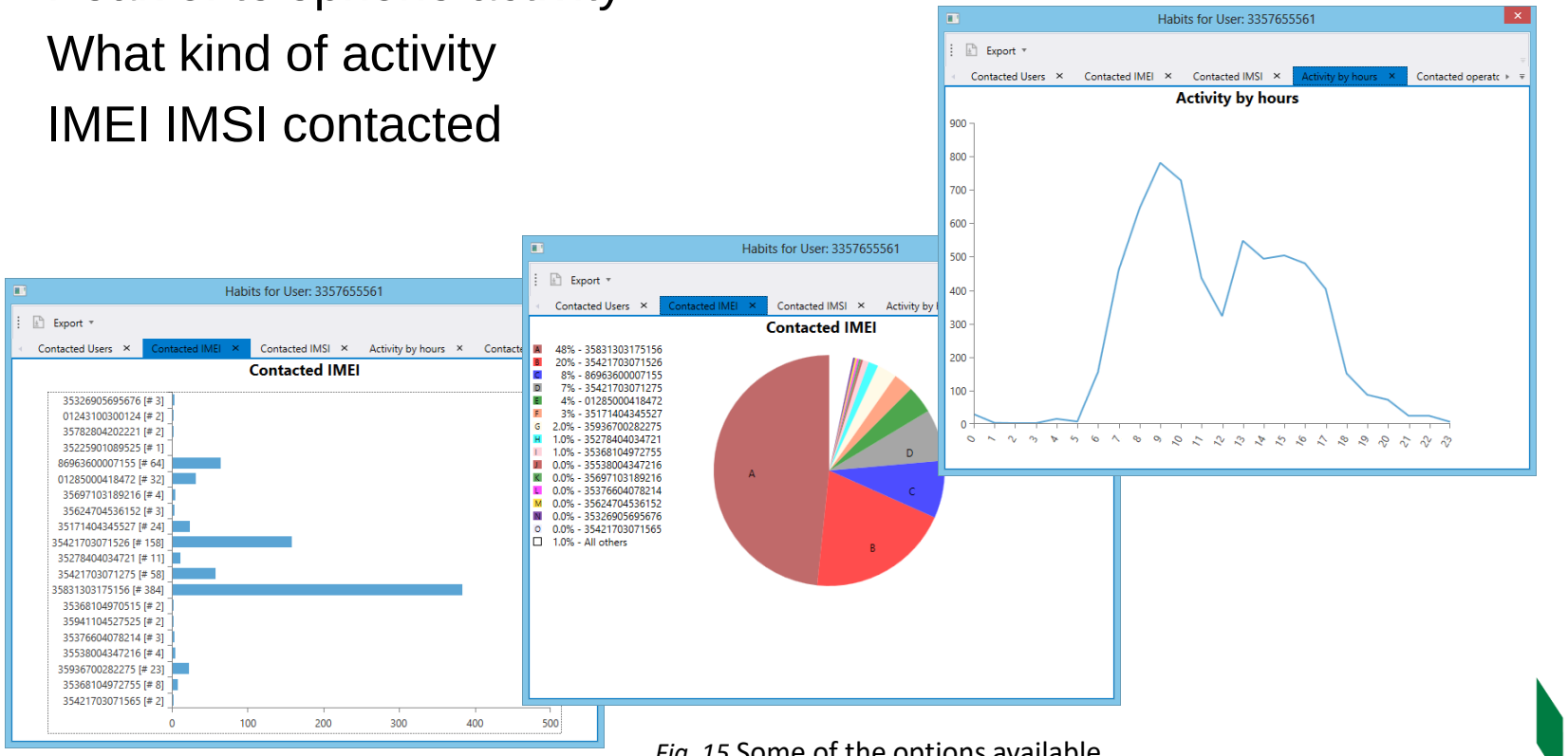


Fig. 15 Some of the options available

UFED, OXYGEN Forensic, XRY reports

The software imports and makes analysis with UFED,OXYGEN forensic and XRY reports.

Research on:

- Pictures (EXIF data)
- WiFi connections
- BTS roaming
- POIs from Map App

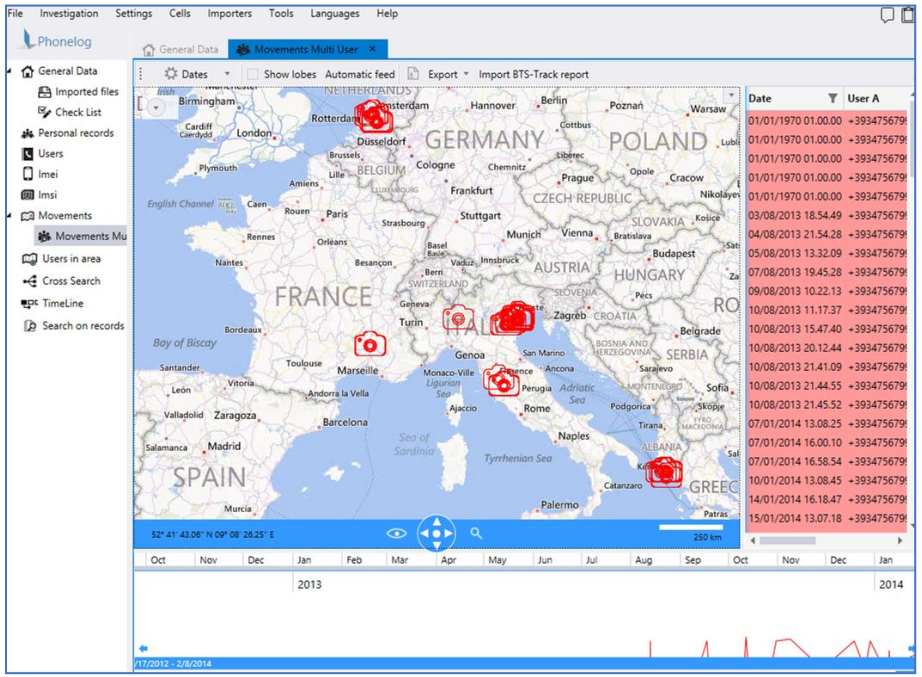


Fig. 16 Overview on UFED report analysis

Lightning-fast investigations

Thanks to SecurCube Phone Log:

- ✔ With a very few clicks, the Investigator can slim down the list of CDR records and check the starting points
- ✔ Fast and intuitive research
- ✔ Wide range of functionalities up to investigators' standards
- ✔ Quickly finds information of a suspicious user, define a profile of suspect
- ✔ Exports graphical results for non-technical personnel
- ✔ Fast growing efficiency of the results obtained